

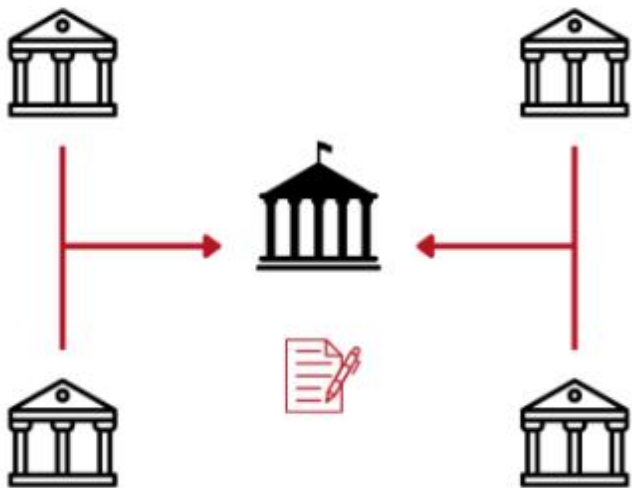
Digital finance & Blockchain Technology



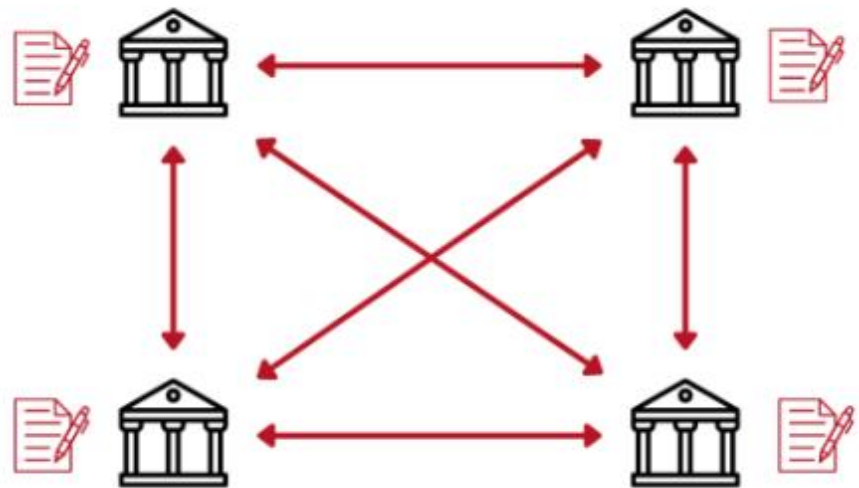
Motivation, Components, and Core Concepts

Motivation Behind Blockchain

Centralised Ledger



Decentralised Ledger



Motivation Behind Blockchain



Lack of trust in centralized systems



Need for transparency and immutability



Reducing intermediaries in transactions

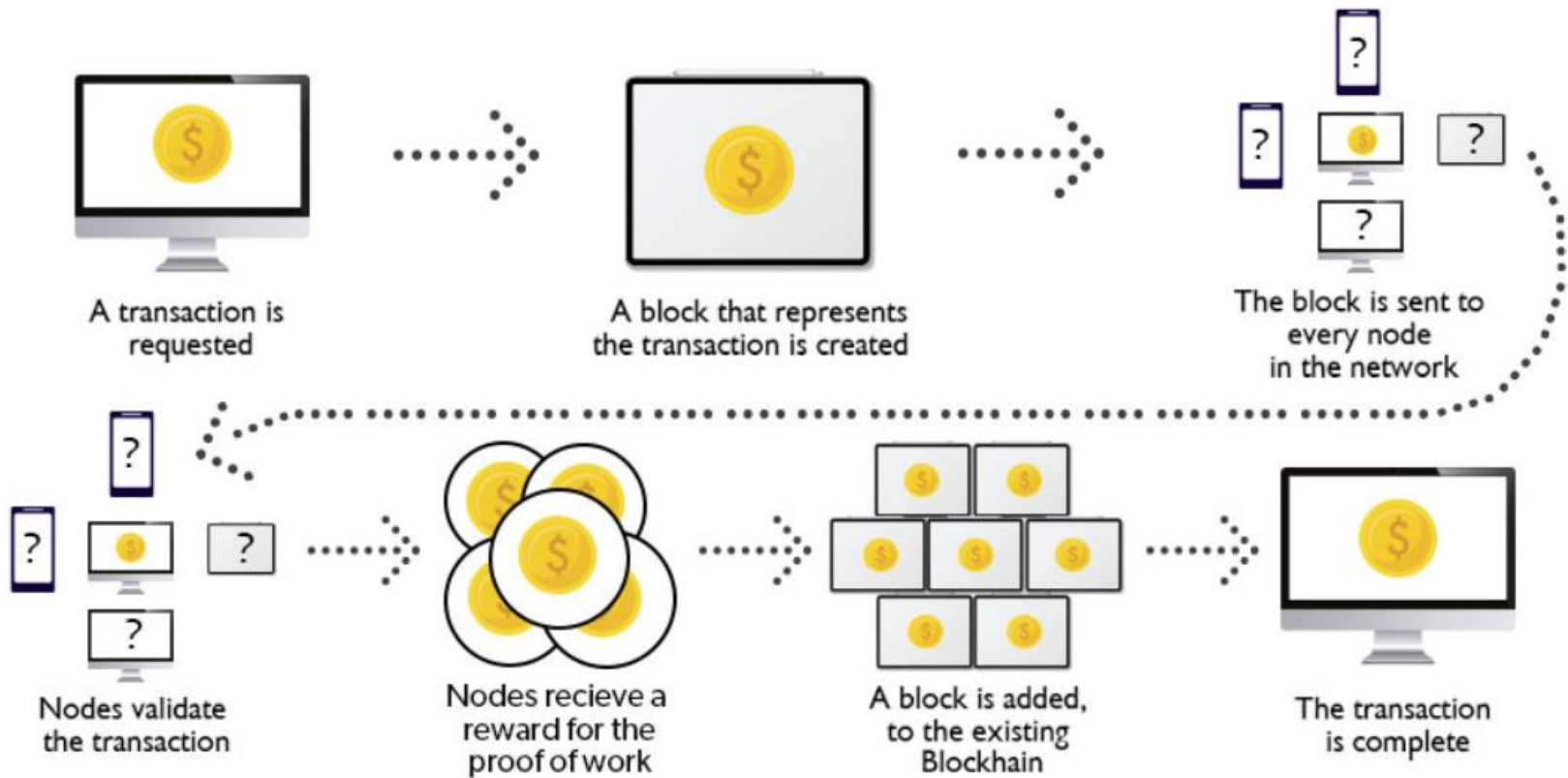


Secure peer-to-peer value transfer



Resistance to fraud and data tampering

Overview of Blockchain



Overview of Blockchain

- Distributed and decentralized digital ledger
- Records transactions in blocks
- Blocks are linked using cryptography
- Maintained by a network of nodes
- Data once added cannot be easily altered

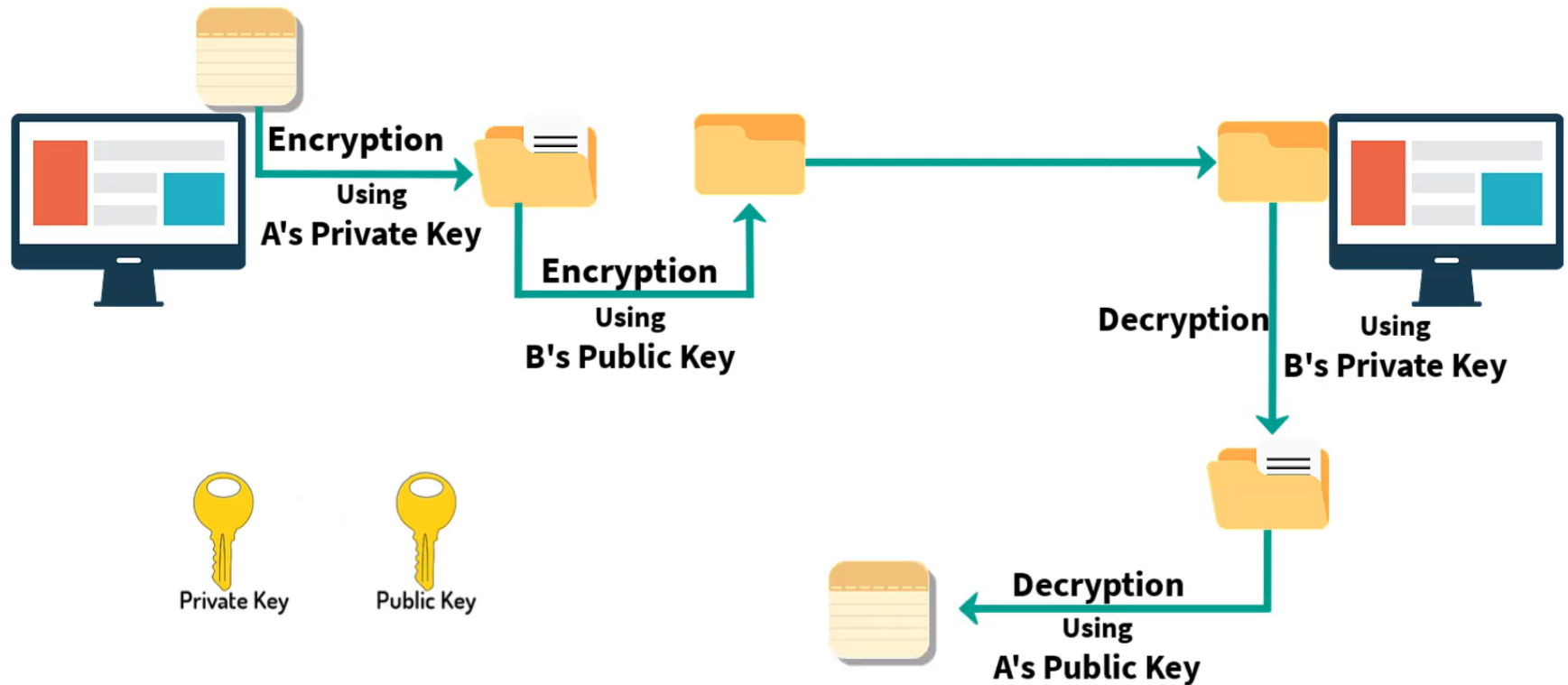
Components of Blockchain

- Digital Signatures
- Hashing
- Consensus Mechanism
- Distributed Ledger
- Network Nodes

Digital Signatures

- Use public key cryptography
- Utilizes both private and public key
- Ensure authentication and non-repudiation
- Verify the sender of a transaction
- Commonly implemented using RSA or ECDSA

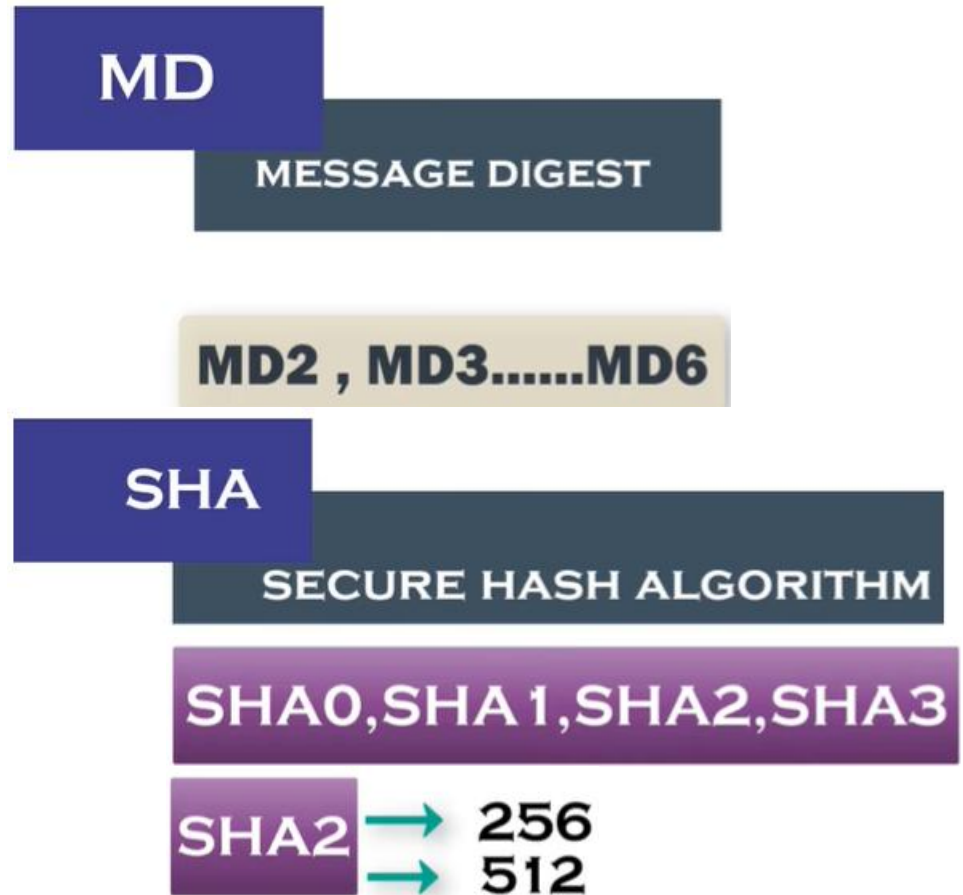
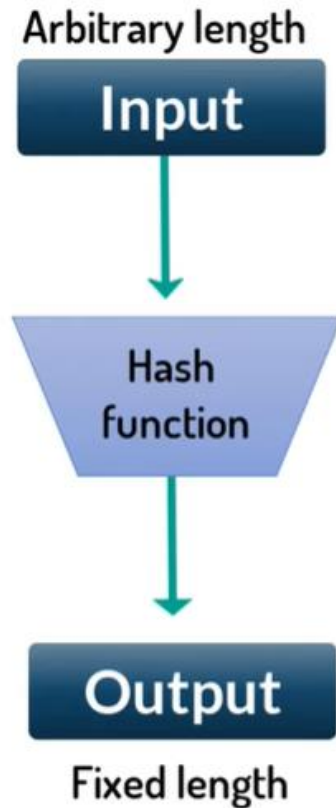
Digital Signatures



Hashing

- Converts input data into fixed-size output
- One-way and collision-resistant
- Links blocks securely
- Common hash function: SHA-256

Hashing



Consensus Mechanism

- Process by which nodes agree on ledger state
- Prevents double spending
- Maintains network security
- Examples: Proof of Work, Proof of Stake

Consensus Mechanism

Proof of Work		VS	Proof of Stake	
	The probability of producing a block is dependent on the amount of computational power			The probability of producing a block is dependent on the amount staked on the network
	Miners receive block rewards and transaction fees to solve complex cryptographic puzzles			Validators receive transaction and priority fees for producing blocks
	Hackers would need to have computers more powerful than 51% of the network to add a malicious block, leading to a 51% attack			Hackers would need to stake more than 51% of staked cryptocurrency on the network, making 51% attacks extremely unlikely

Cryptocurrency (Overview)

- Digital or virtual currency
- Uses cryptography for security
- Operates without central authority
- Examples: Bitcoin, Ethereum



cryptocurrencies are
immune to counterfeiting

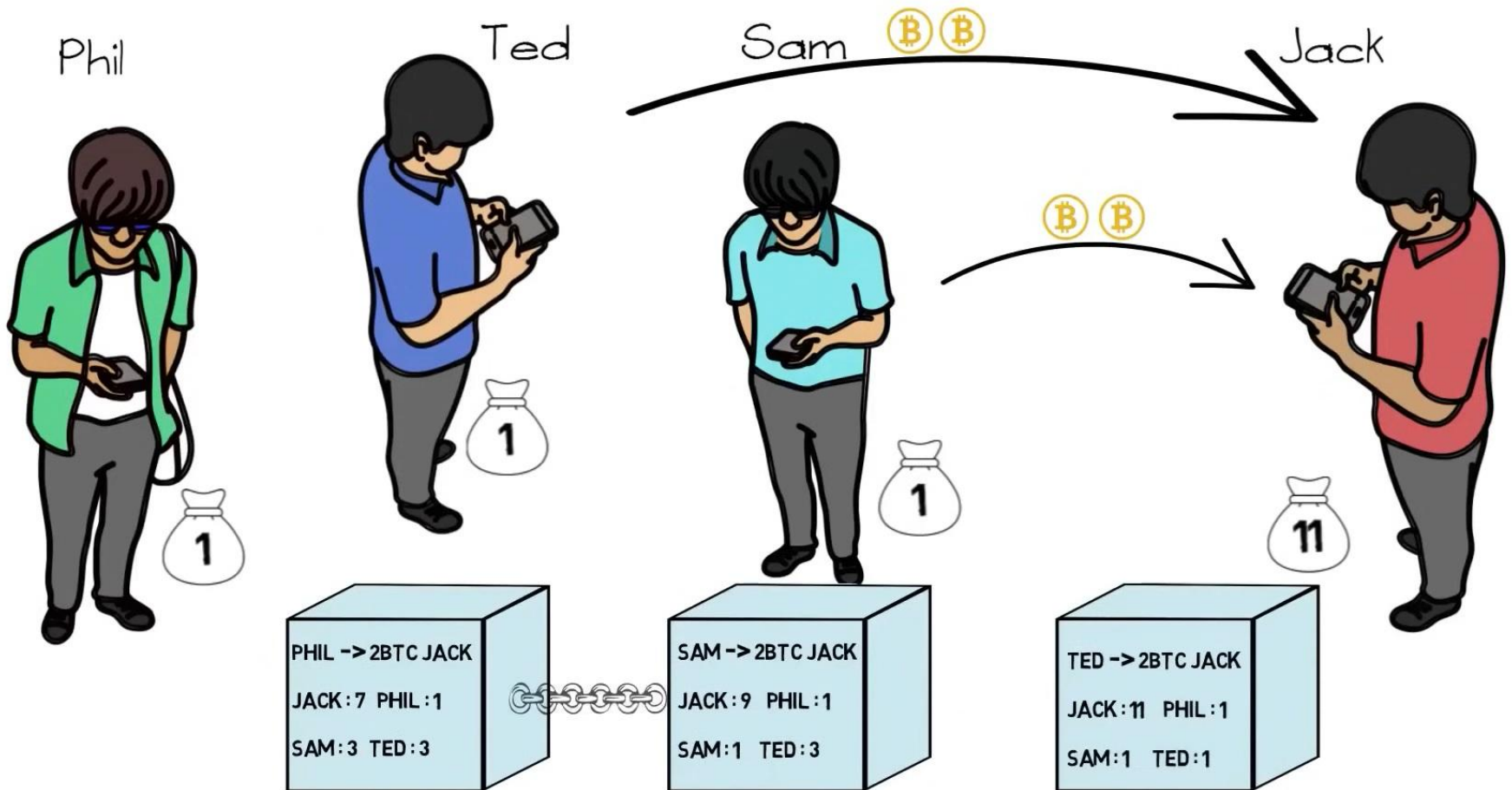


don't require a
central authority

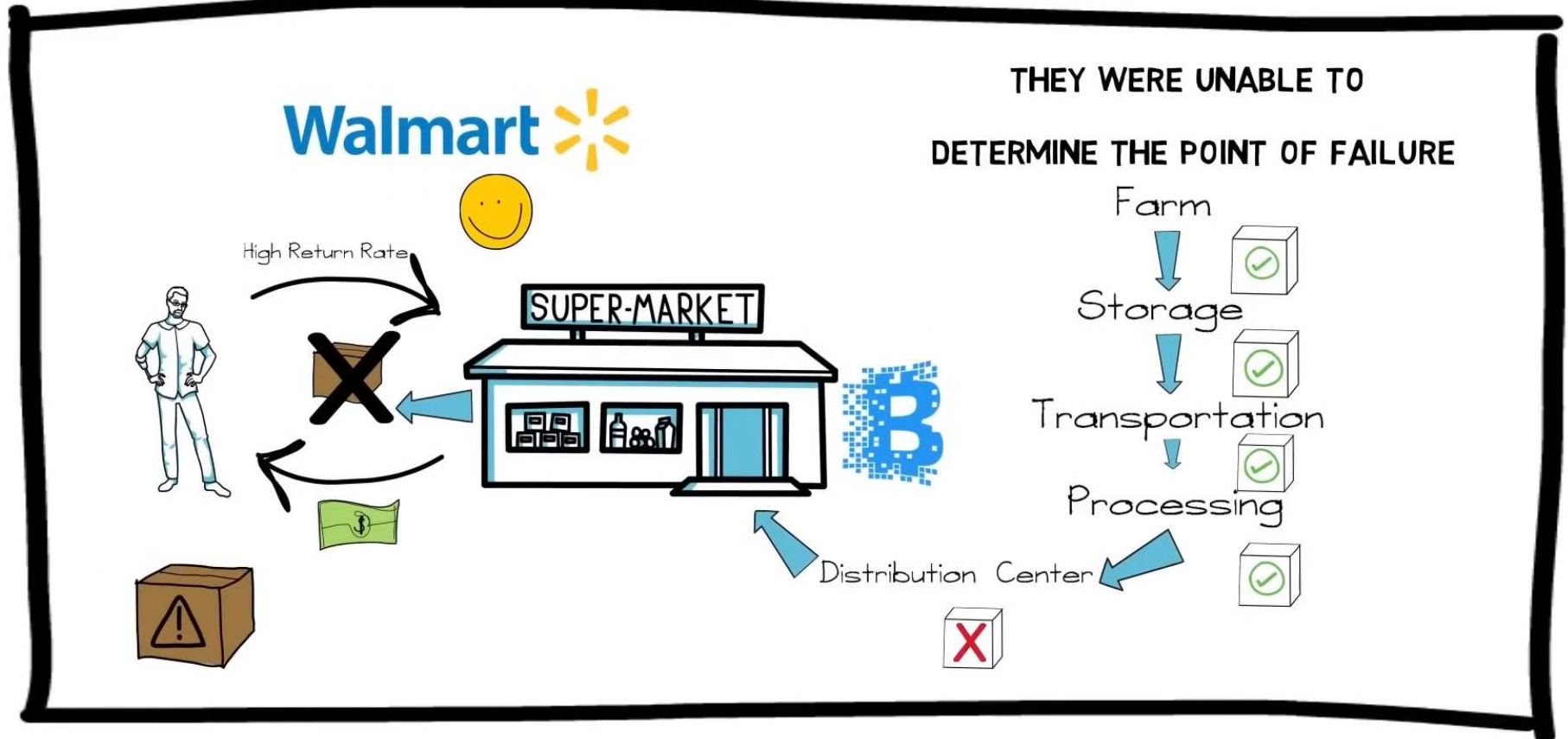


protected by strong and
complex encryption algorithms

Cryptocurrency (Overview)

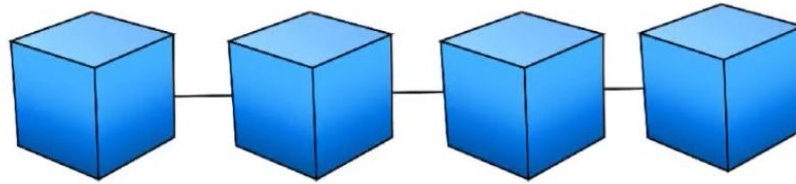


Cryptocurrency (Overview)



HELPING WALMART TO IDENTIFY THE PROBLEM AREAS AND FIXING THEM

Cryptocurrency (Overview)

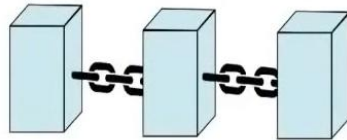


ALL OF THIS IS MADE POSSIBLE WITH THE HELP OF BLOCKCHAIN TECHNOLOGY

BLOCKCHAIN CAN BE DESCRIBED AS



collection of records



linked with each other

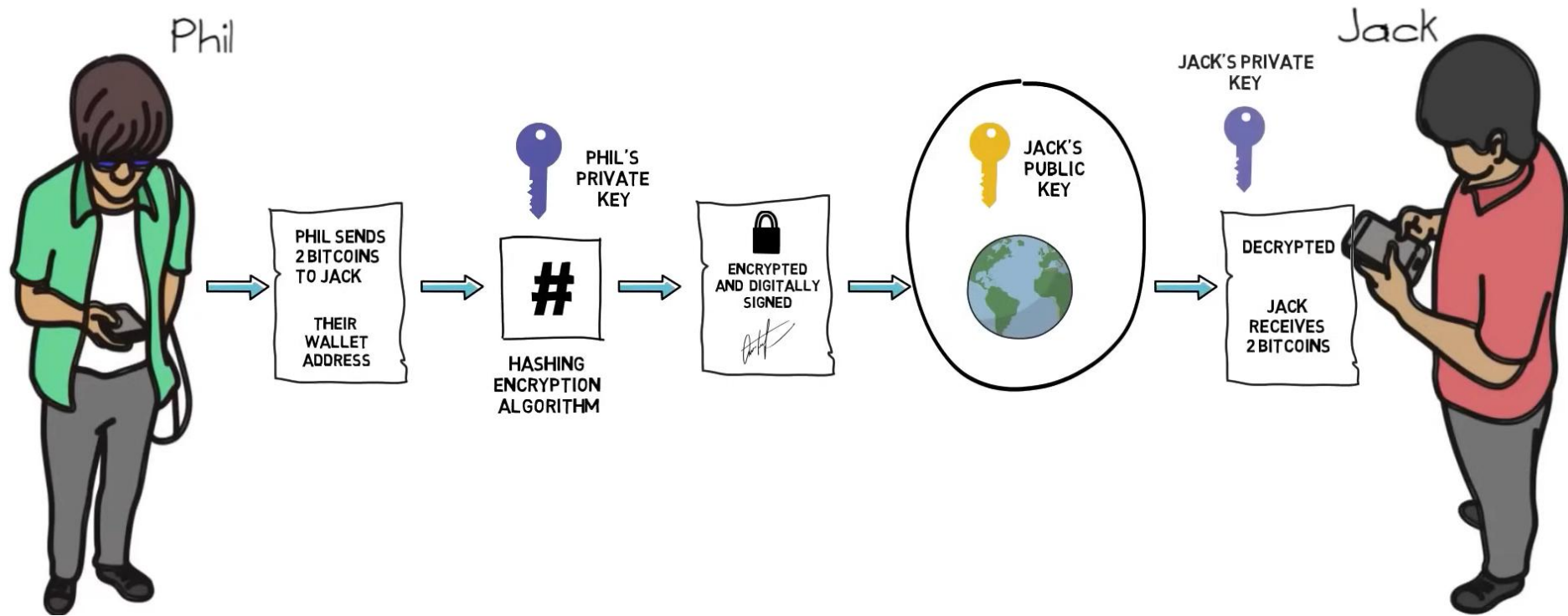


strongly resistant
to alteration

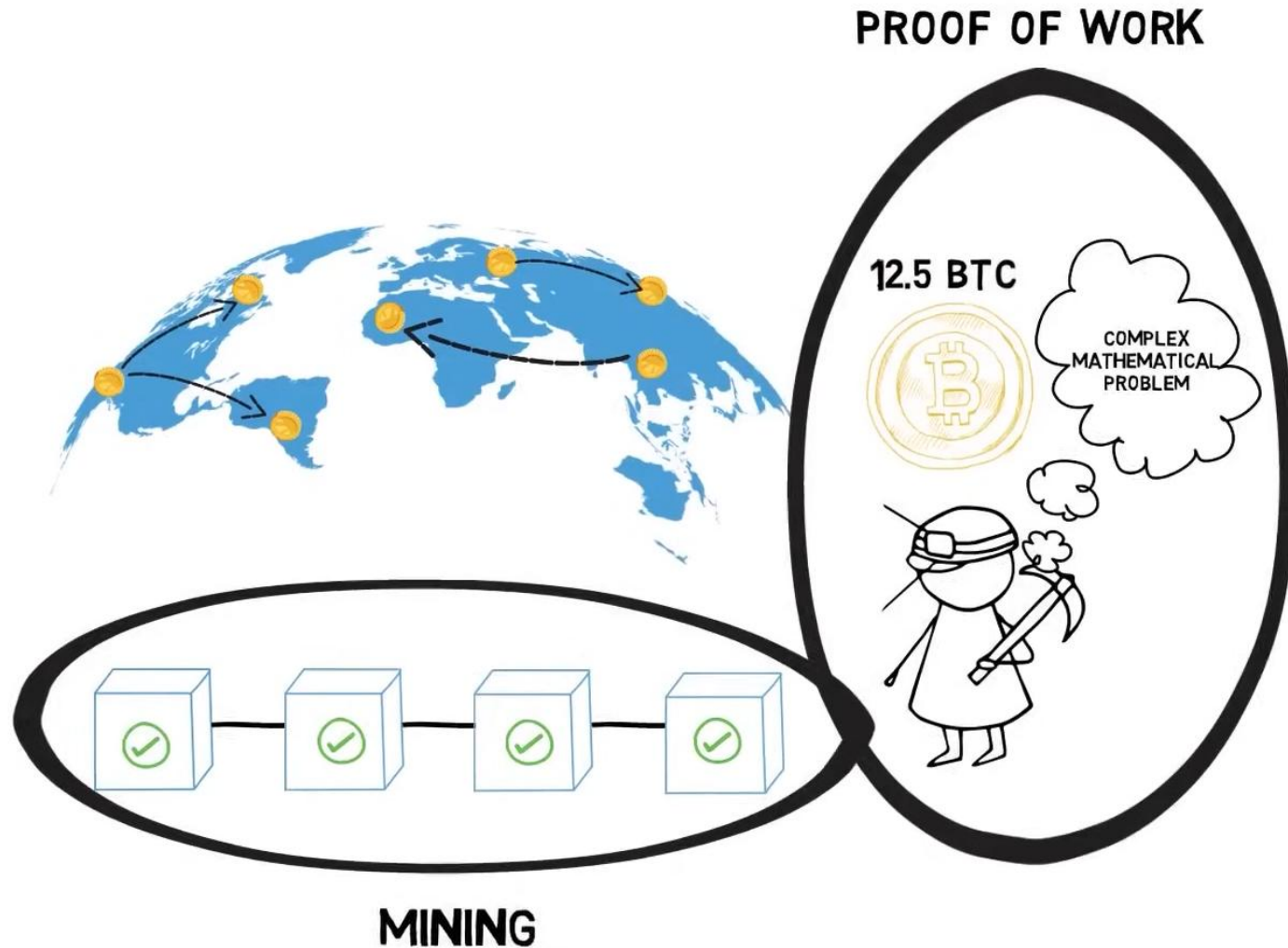


protected using
cryptography

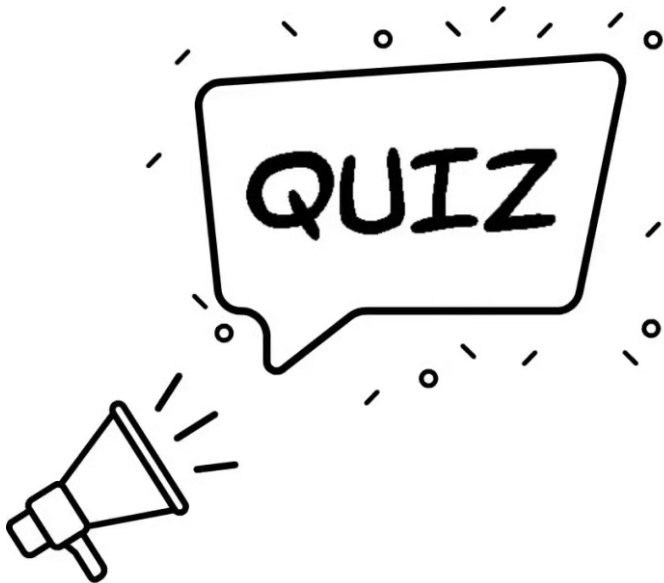
Cryptocurrency (Overview)



Cryptocurrency (Overview)



**WHAT IS THE CONCEPT OF BLOCKCHAIN THAT ENSURES DATA
CANNOT BE ALTERED BY ANY OF THE USERS WITHIN THE NETWORK?**



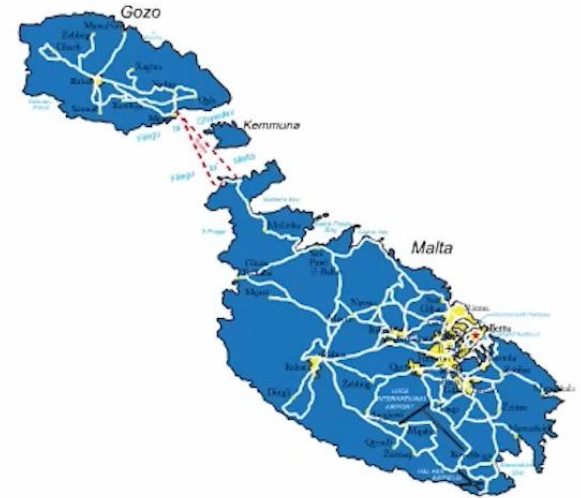
- A: Public Distributed Ledger
- B: Proof of Work
- C: Proof of Stake
- D: Hash encryption

Cryptocurrency (Overview)

In 2018,



BLOCKCHAIN ISLAND



 **BINANCE**

 **OKEX**



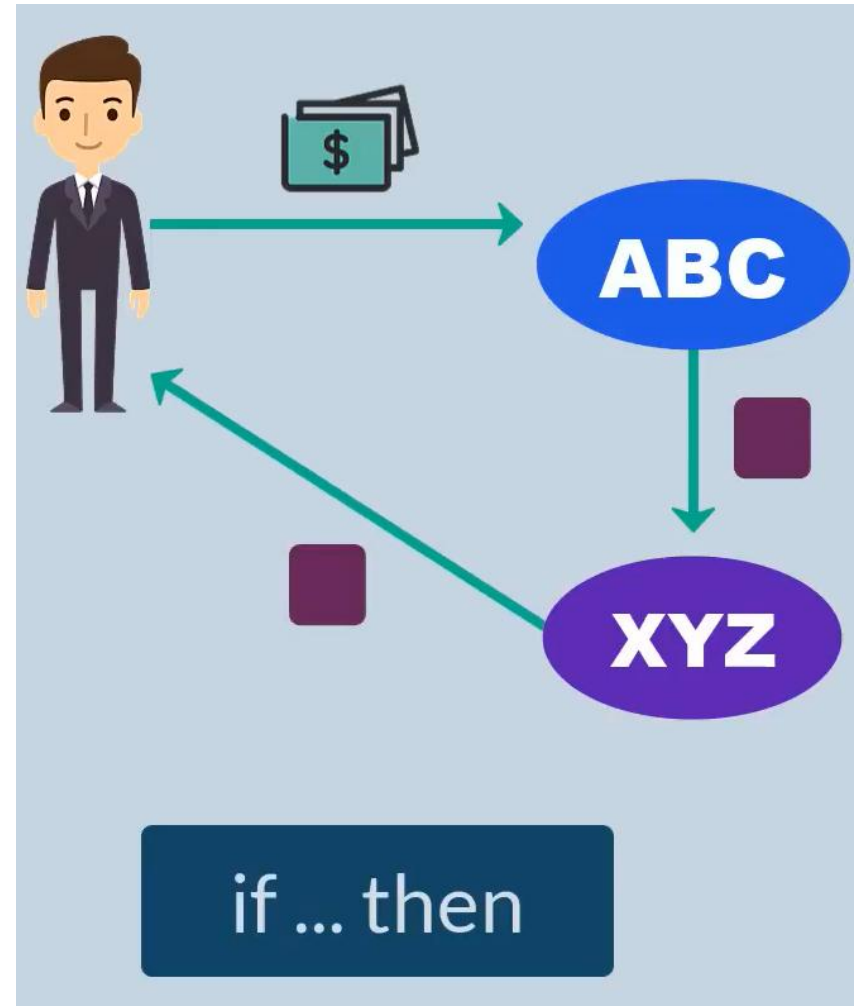


Which Hashing algorithm does cryptocurrency use?

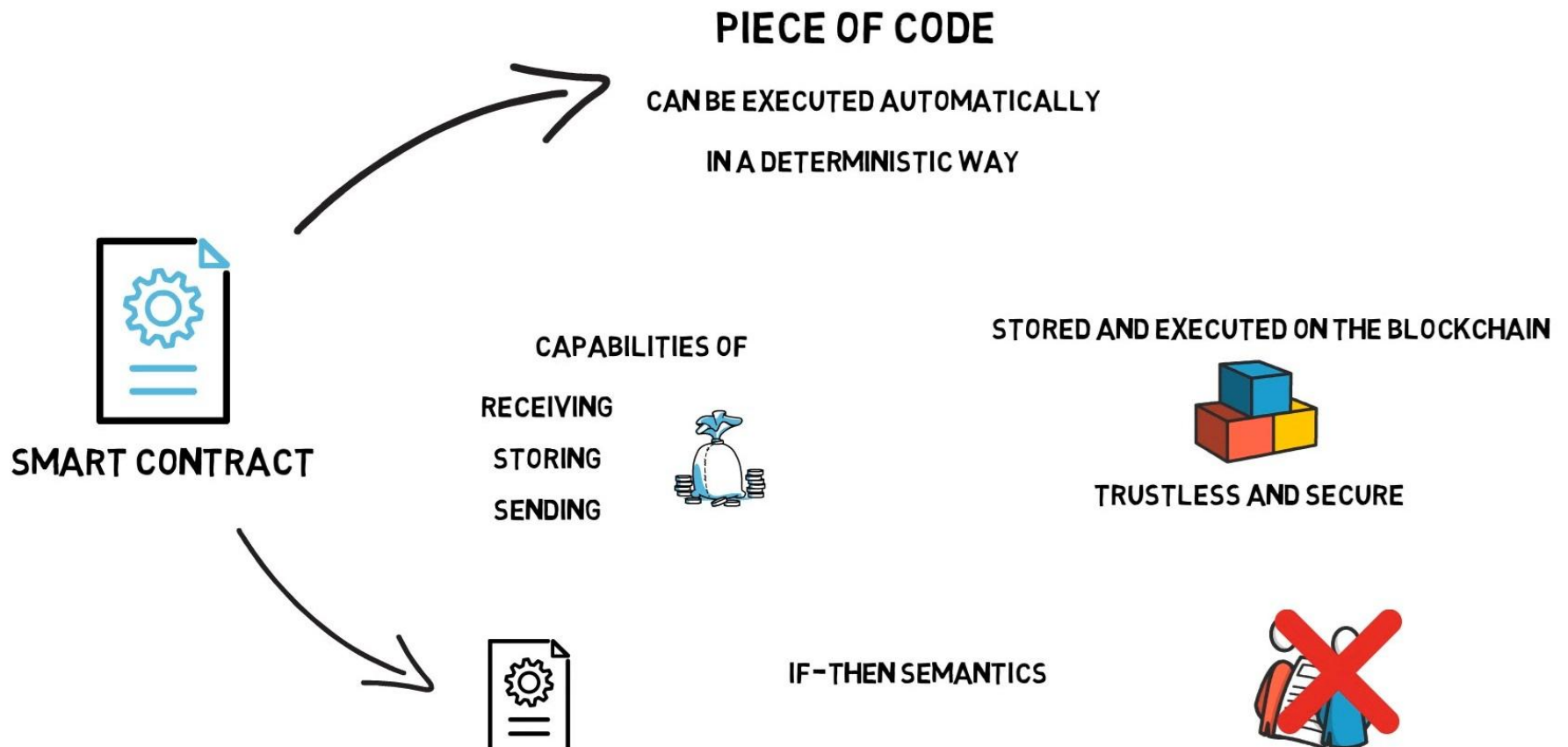
- a) Scrypt
- b) Ethash
- c) SHA256
- d) None of the above

Smart Contracts (Overview)

- Self-executing contracts with coded rules
- Automatically enforce agreements
- Run on blockchain platforms like Ethereum
- Reduce need for intermediaries



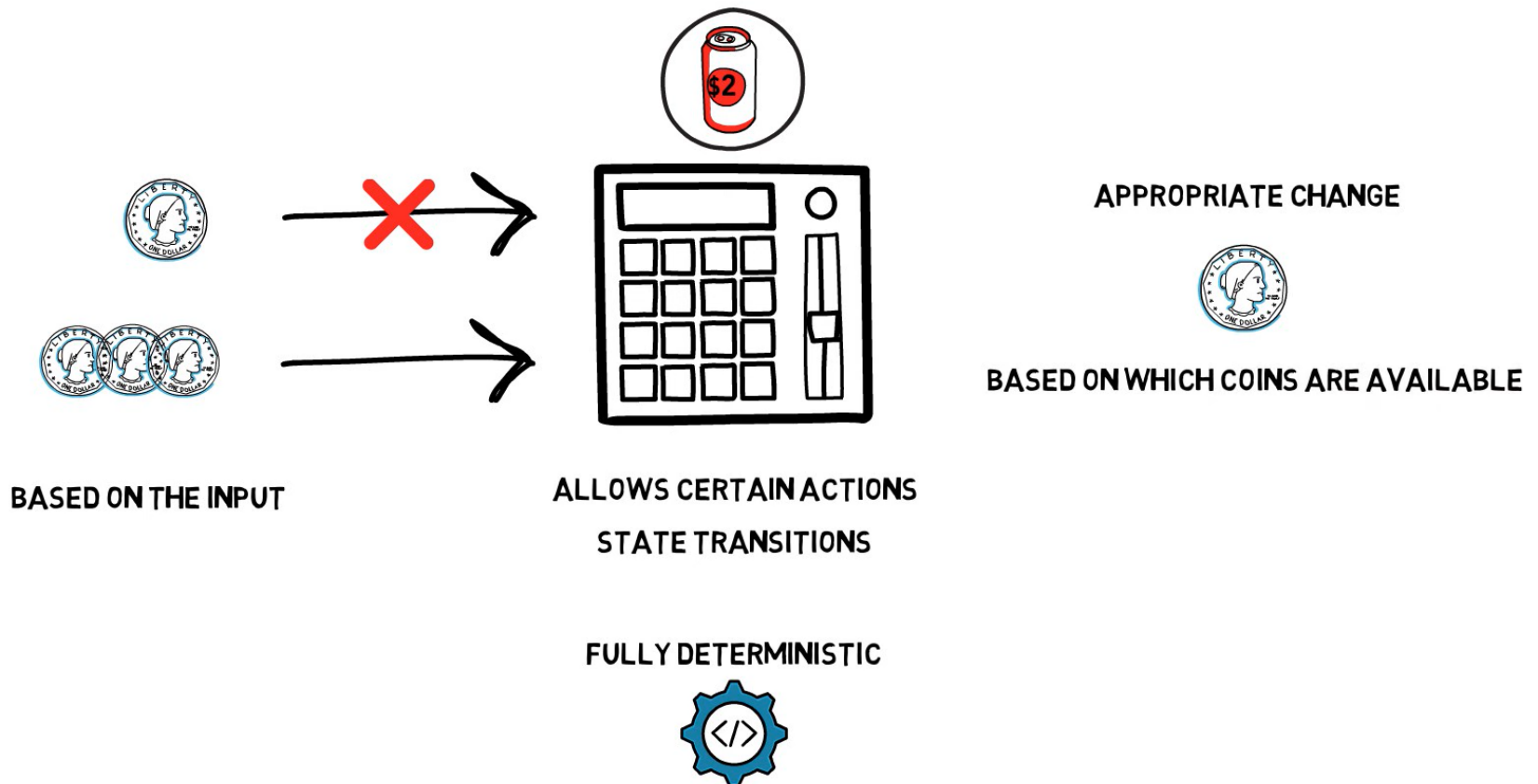
Smart Contracts (Overview)



Smart Contracts (Overview)

VENDING MACHINE

ANALOGY TO A SMART CONTRACT



Smart Contracts (Overview)

THE MOST POPULAR GENERAL-PURPOSE SMART CONTRACT PLATFORM



ETHEREUM

BLOCKCHAIN

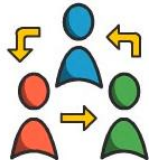
SUPPORTS SMART CONTRACTS



IMMUTABLE

ONCE DEPLOYED CANNOT BE MODIFIED

DECENTRALIZED



EOS



SOLIDITY

CREATES CERTAIN RISKS

Smart Contracts (Overview)

"SMART CONTRACT"



COINED BY A WELL-KNOWN CRYPTOGRAPHER

NICK SZABO

1990s



NOT THE MOST SELF-EXPLANATORY

COMMONLY USED IN THE BLOCKCHAIN INDUSTRY



Smart Contracts (Overview)

THE MOST POPULAR GENERAL-PURPOSE SMART CONTRACT PLATFORM



ETHEREUM

BLOCKCHAIN

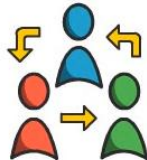
SUPPORTS SMART CONTRACTS



IMMUTABLE

ONCE DEPLOYED CANNOT BE MODIFIED

DECENTRALIZED



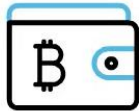
SOLIDITY

CREATES CERTAIN RISKS

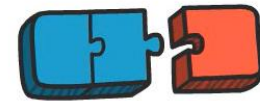
Smart Contracts (Overview)

USE CASES

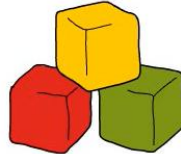
PAYMENTS



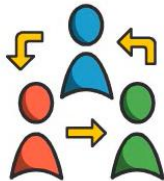
SUPPLY CHAIN



DAPPS



DECENTRALIZED FINANCE



CROWDFUNDING



Smart Contracts (Overview)

THE MOST POPULAR GENERAL-PURPOSE SMART CONTRACT PLATFORM



ETHEREUM

BLOCKCHAIN

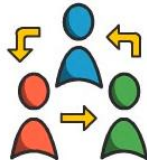
SUPPORTS SMART CONTRACTS



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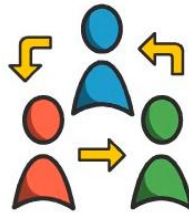


SOLIDITY

CREATES CERTAIN RISKS

Smart Contracts (Overview)

DECENTRALIZED FINANCE (DEFI)



DECENTRALIZED STABLE COIN



PEGGED TO



MAKERDAO

AUTOMATED LIQUIDITY PROVISIONING



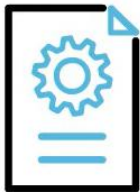
PROVIDE LIQUIDITY SWAP TOKENS



PERMISSIONLESS
DECENTRALIZED

Smart Contracts (Overview)

OTHER EXAMPLES



RIDE-SHARING



APARTMENT RENTAL

CHARITY



FUND



NO INTERMEDIARIES



AND MUCH MORE?

Non-Fungible Tokens (NFTs)

- Unique digital assets on blockchain
- Represent ownership of digital or physical items
- Not interchangeable like cryptocurrencies
- Used in art, gaming, music, and collectibles

